

Aerodynamic Coefficients Prediction Presentation

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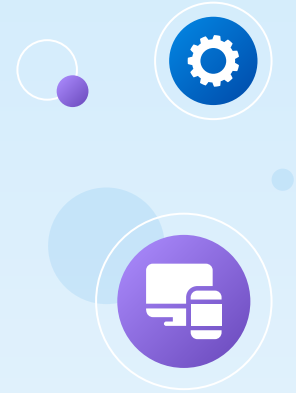
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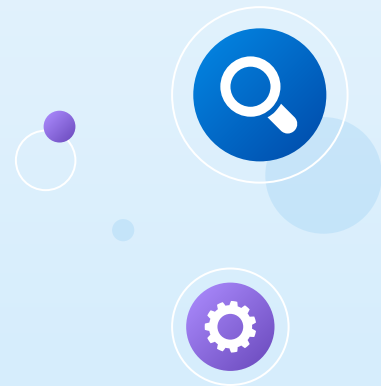
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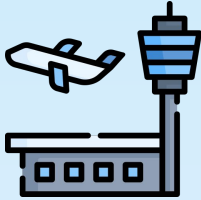


01

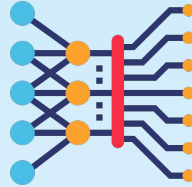
Introduction →



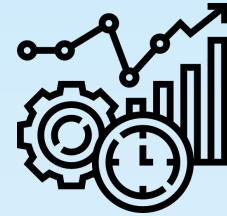
Introduction



Aviation industry has started to embrace data science to revolutionize aerodynamic analysis with machine learning.



The integration of machine learning has augmented the traditional top-down approach, leading to improved understanding of previously unsolved aerodynamic challenges.



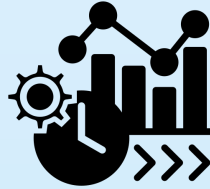
Machine learning enables accurate and efficient optimization prediction in all areas where it can be applied.



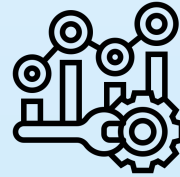
Motivation



Highly optimized aerodynamic designs to meet the increasing demand for new passenger aircraft while adhering to stringent emission targets.



Machine learning enables efficient airfoil optimization, enhancing aircraft performance and reducing costs in the aviation industry.



Aerodynamic shape optimization that characterized by reduced drag and high lift-to-drag ratios .

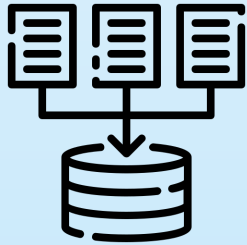
02

Objectives →

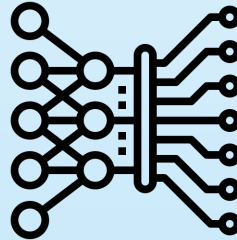


Primary Goal

The main objective of this project is to develop a machine learning tool that accurately classifies the airfoil with the most optimal lift-to-drag ratio within the Reynolds Number range of 50,000–1,000,000 and angle of attack range of -19.5 to 19 degrees.



Accumulate data

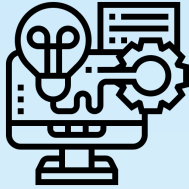


Use Machine
learning



To predict,
compare and
generate

Scope



Test Machine Learning Models

- Linear Regression
- Lasso
- Ridge
- ElasticNet
- Bayesian Ridge
- Random Forest Regression
- Support Vector Regression
- Multi-Layer Perceptron



Data

- Access airfoil data from XFOIL analysis results published on airfoiltools.com.
- Focus on aerodynamic characteristics at set effective angles of attacks of (-19.5 to 19 degrees) and Reynolds numbers of (50,000-1,000,000)



Model Ranking Criteria

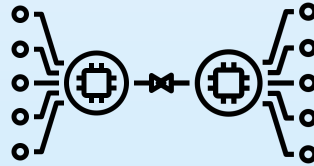
- Selecting the model with the highest R^2 as the most optimal for predicting C_l/C_d
- Achieve a target R^2 score of 0.80 and above, indicating that the model explains more than 80% of the variance in C_l/C_d .

Scope (contd.)



Airfoils used:

- NACA 0012
- NACA 1412
- NACA 2412
- NACA 4412
- NACA 6412

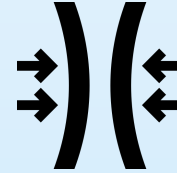


Parameters Input:

- Reynolds Number (Re)
- Angle of attack (AoA)

Parameters Output:

- Optimal Airfoil
- Maximum C_l/C_d
- R^2 score



Constraints:

- Reynolds Number: [50000,1000000]
- Angle of attack: [-19.5, 19] degrees

03

Methodology →



Methodology



Source Data

The study used publicly available databases of airfoil geometries and performance data, which were analyzed using XFOIL software to simulate the aerodynamic properties of airfoils and predict their performance.

The Idea

The methodology used in this study involved a comprehensive approach that combined computational simulations, and machine learning techniques to develop a predictive model for the aerodynamic performance of different airfoils

Source Data

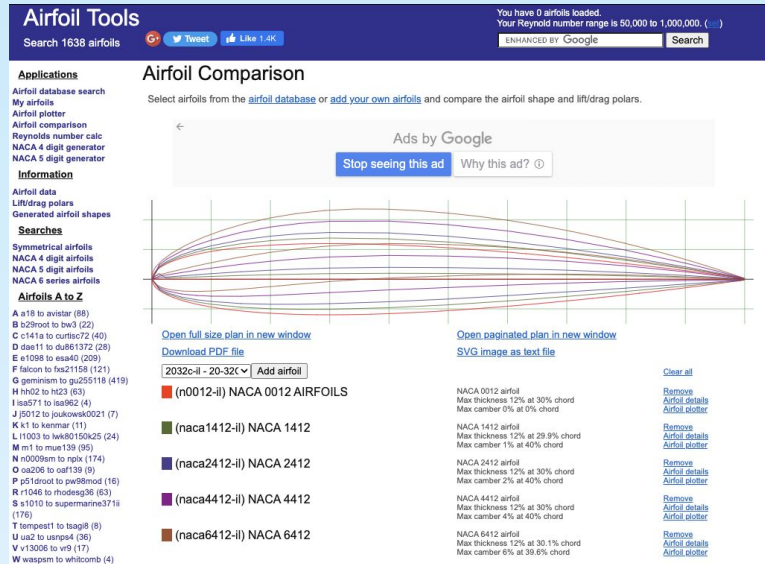


Figure 1: Data Collection from [airfoiltools.com](http://airfoiltools.com/compare/index)

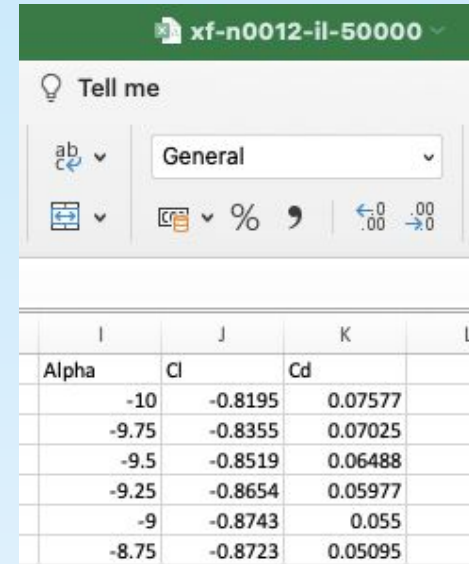


Figure 2: Raw Data from Source

Collects airfoil data from published source, *airfoiltools.com*, and extract the relevant data for the analysis.

Source Data

```
import os
import csv

# Define the folder containing the CSV files
folder_path = '/Users/jarkrunglekriangkrai/ISE/Year 4/Senior Project/airfoils'

# Define the output CSV file path
output_file_path = '/Users/jarkrunglekriangkrai/ISE/Year 4/Senior Project/Book1.csv'

# Loop over each file in the folder
with open(output_file_path, 'w', newline='') as output_file:
    writer = csv.writer(output_file)

    for filename in os.listdir(folder_path):
        # Only process files with the ".csv" extension
        if filename.endswith('.csv'):
            # Construct the full path to the file
            filepath = os.path.join(folder_path, filename)

            # Open the file for reading
            with open(filepath, 'r', newline='') as csvfile:
                # Create a CSV reader object
                reader = csv.reader(csvfile)

                # Skip the first 10 rows
                for i in range(11):
                    next(reader)

                # Read the remaining rows into a list
                remaining_rows = list(reader)

                # Add a new column with the file name
                for row in remaining_rows:
                    row.append(filename)

                # Write the remaining rows to the output file
                writer.writerows(remaining_rows)
```



	A	B	C	D	E	F	G
1	Alpha	Reynolds	Airfoil	Cl/Cd			
2	-19.25	500000	naca0012				
3	-19	500000	naca0012				
4	-18.75	500000	naca0012				
5	-18.5	500000	naca0012				
6	-18.25	500000	naca0012				
7	-18	500000	naca0012				
8	-17.75	500000	naca0012				
9	-17.5	500000	naca0012				
10	-17.25	500000	naca0012				
11	-17	500000	naca0012				
12	-16.75	500000	naca0012				
13	-16.5	500000	naca0012				
14	-16.25	500000	naca0012				
15	-16	500000	naca0012				
16	-15.75	500000	naca0012				
17	-15.5	500000	naca0012				
18	-15.25	500000	naca0012				
19	-15	500000	naca0012				
20	-14.75	500000	naca0012				
21	-14.5	500000	naca0012				
22	-14.25	500000	naca0012				
23	-14	500000	naca0012				
24	-13.75	500000	naca0012				
25	-13.5	500000	naca0012				
26	-13.25	500000	naca0012				
27	-13	500000	naca0012				
28	-12.75	500000	naca0012				

Sort

By color:

Filter

By color:

Choose One

Search

- ☒ (Select All)
- ☒ naca0012
- ☒ naca1412
- ☒ naca2412
- ☒ naca4412
- ☒ naca6412

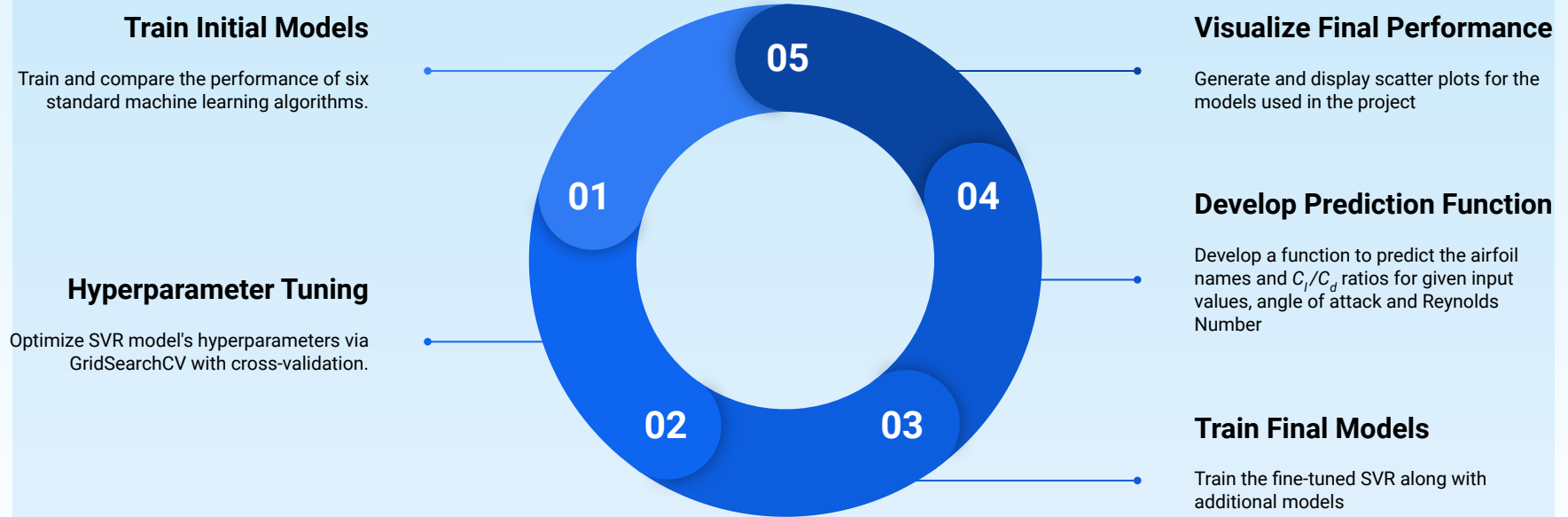
☒ Auto Apply

Figure 3: Python Code Used for Data Preparation

Figure 4: Raw Data After Cleaning and Transformation

Raw data are converted, combined, and prepared into analysis-ready format using file and csv manipulation using Python and Excel.

Overall Flow



01 Train Initial Models

Models

- Support Vector Regression
- Linear Regression
- Lasso
- Ridge
- ElasticNet
- Bayesian Ridge

Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 score

Value

Generally, it is a good idea to train multiple models and these initial standards models can give a baseline understanding and provide a starting point for further refinement

01 Train Initial Models: Summary

Summary Points

- Test size = 30%
- SVR (C = 0.1, Epsilon = 0.1, Kernel = "Linear")
- Lasso (Alpha = 0.1)
- Ridge (Alpha = 0.1)
- ElasticNet (Alpha = 0.1, L1 Ratio = 0.5)
- Linear Regression
- Bayesian Ridge
- C: the penalty parameter of the error term
- Epsilon: the tolerance for error
- Kernel: the type of kernel to use in SVM algorithm
- Alpha: responsible for preventing the overfitting by adding a penalty
- L1 Ratio: the balance between Lasso and Ridge

01 Train Initial Models: Result

Key Points

- Similar performance
- Fairly high MAE and MSE
- R^2 score not achieving the required threshold of 0.80

Model	MAE	MSE	R^2
SVR	24.12	968.11	0.50
Linear Regression	24.24	932.65	0.52
Lasso	24.24	932.99	0.52
Ridge	24.24	932.65	0.52
ElasticNet	24.33	937.85	0.51
Bayesian Ridge	24.24	932.70	0.52

Table 1: Performance Matrix for 6 Initial Models

02 Hyperparameter Tuning

Manual Search Method

Use the hit and trial process and can find the best hyperparameters

Grid Search Method

Use a different combination of all the specified hyperparameters and their values. Calculates the performance for each combination and then select the best value for the hyperparameters

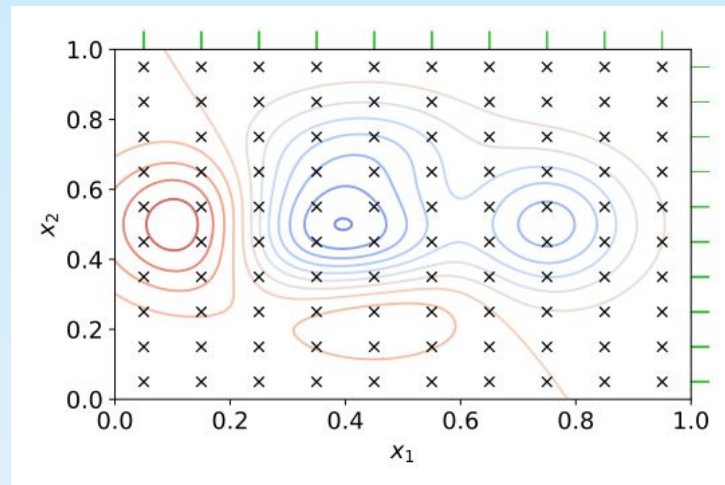


Figure 5: Grid Search Across Two Parameters
<https://www.analyticsvidhya.com/blog/2021/06/tune-hyperparameters-with-gridsearchcv/>

02 Hyperparameter Tuning: Result

```
# Define hyperparameters grid search
param_grid = {
    'C': [0.1, 1, 10],
    'epsilon': [0.1, 0.2, 0.3],
    'kernel': ['linear', 'poly', 'rbf']
}

# Perform grid search with cross-validation
grid_search = GridSearchCV(svr,
    param_grid=param_grid, cv= 5, n_jobs=-1)
grid_search.fit(X_train, y_train)

# Print best hyperparameters
print('Best hyperparameters:',
    grid_search.best_params_)

Output: Best hyperparameters:
{'C': 10, 'epsilon': 0.3, 'kernel': 'rbf'}
```

Parameters

- C: the penalty parameter of the error term
- Epsilon: the tolerance for error
- Kernel: the type of kernel to use in the SVM algorithm

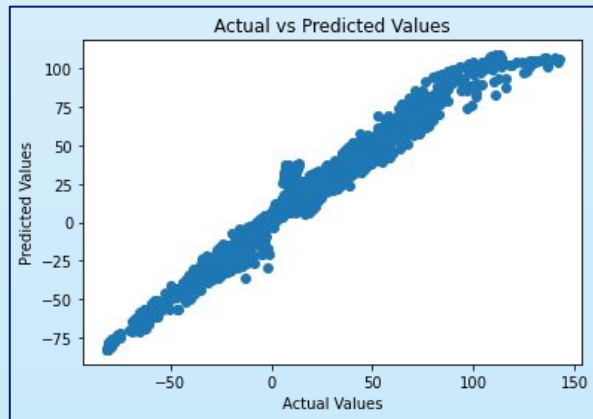


Figure 6: Model Performance with the best hyperparameters

03 Train Final Models

Models

- Linear Regression
- Random Forest
- Support Vector Regression (Hyperparameters)
- Multi-layer Perceptron

Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 score

Value

Fine-tuning the model can result in increase in model performance. The flexible models will always be a good choice in finding the best model

03 Train Final Models: Summary

Summary Points

- Test size = 20%
- SVR (C = 10, Epsilon = 0.1, Kernel = "Linear")
- Random Forest (n_estimators = 100)
- MLP (hidden_layer_sizes = (100,100), max_iter = 10,000)
- Linear Regression

03 Train Final Models: Result

Key Points

- SVR and MLP performed well
- Fairly low MAE and MSE
- R^2 score achieving the required threshold of 0.80

Model	MAE	MSE	R^2
Linear Regression	12.82	489.56	0.23
Random Forest	6.99	146.32	0.32
SVR (Fine-tuned)	11.37	244.21	0.86
MLP	11.46	232.00	0.87

Table 2: Performance Matrix for 4 Final Models

04 Develop Prediction Function

Input Parameters

- Angle of attack
- Reynolds Number

Output

- Model
- Airfoil NACA number
- C_l/C_d

```
predict(5, 500000)
```

```
output:
```

Model	Airfoil	C_l/C_d
Linear regression	naca2412	28.786549106390446
Random forest	naca1412	80.69676323052141
Support vector regression	naca1412	78.2022008592873
Multi-layer perceptron	naca1412	83.47611322360702

Table 3: Prediction at 5 degrees AOA and 500,000 Re

05 Visualize Final Performance

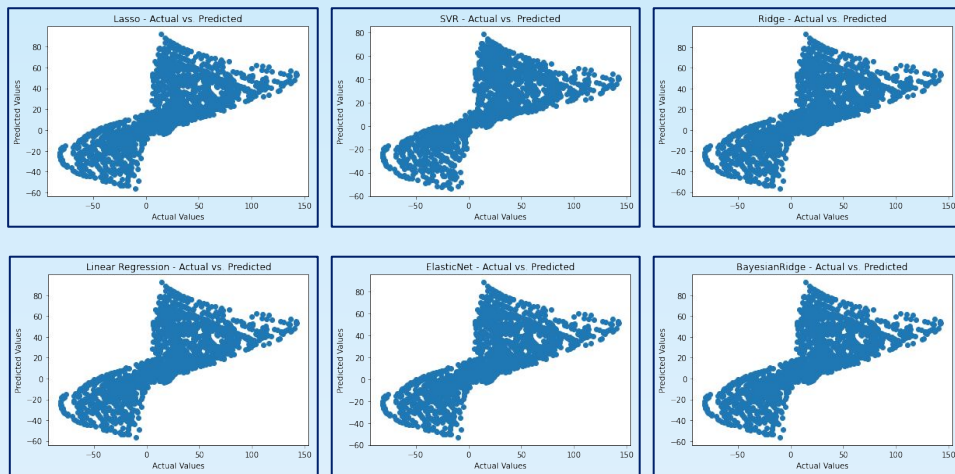


Figure 7: Performance Plot of 6 Initial Models

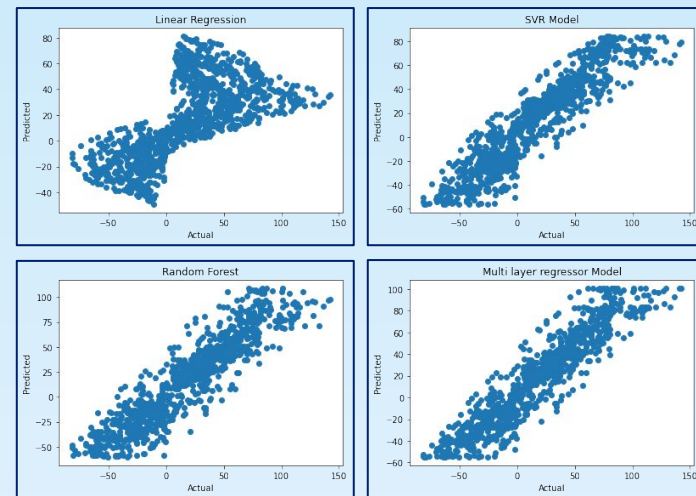


Figure 8: Performance Plot of 4 Final Models

04

Results →



Results

Model	MAE	MSE	R ²
SVR	24.12	968.11	0.50
Linear Regression	24.24	932.65	0.52
Lasso	24.24	932.99	0.52
Ridge	24.24	932.65	0.52
ElasticNet	24.33	937.85	0.51
Bayesian Ridge	24.24	932.70	0.52

Table 4: Performance Matrix of 6 Initial Models

Model	MAE	MSE	R ²
SVR (Fine-tuned)	11.37	244.21	0.86
Linear Regression	12.82	489.56	0.23
MLP	11.46	232.00	0.87
Random Forest	6.99	146.32	0.32

Table 5: Performance Matrix of 4 Final Models

Output

- 72% increase in R² score with fine-tuned SVR
- 47% decrease in MAE with fine-tuned SVR
- 75% decrease in MSE with fine-tuned SVR

Results (contd.)

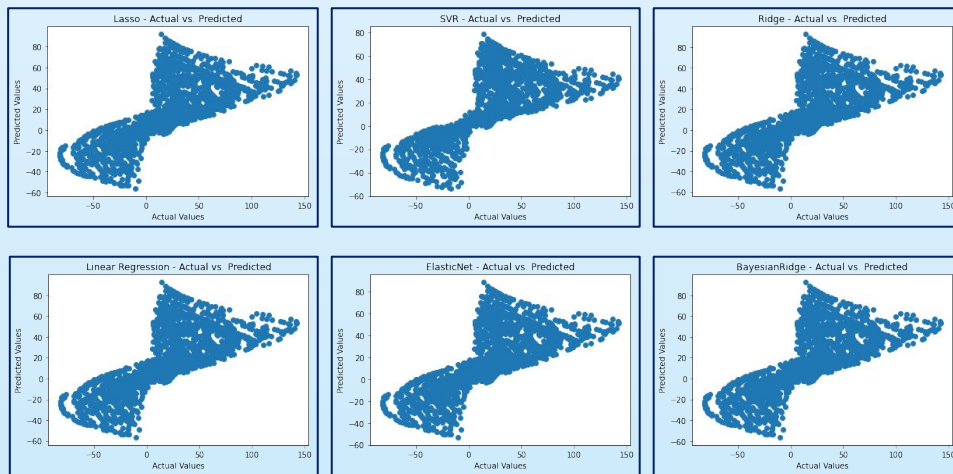


Figure 9: Performance Plot of 6 Initial Models

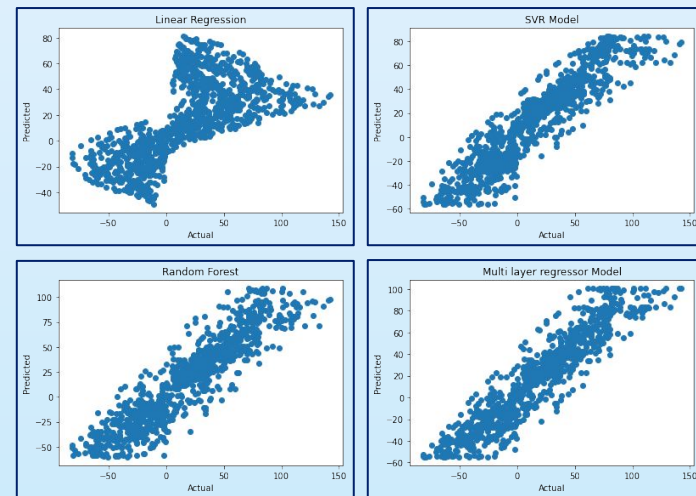


Figure 10: Performance Plot of 4 Final Models

Results: Prediction Function

Input Parameters

- Angle of attack
- Reynolds Number

Output

- Model
- Airfoil NACA number
- C_l/C_d

```
predict(3.9,800000)
```

```
output:
```

Model	Airfoil	C_l/C_d
Linear regression	naca1412	28.739215770120534
Random forest	naca2412	105.24138368293652
Support vector regression	naca2412	75.68073066521814
Multi-layer perceptron	naca2412	98.93913985572546

Table 6: Prediction at 3.9 degrees AOA and 800,000 Re

05

Discussions →



Discussions



Model Performance

The analysis of various regression models revealed a varying degree of performance when applied to the airfoil dataset. The performance was assessed using several metrics such as R^2 score, Mean Absolute Error (MAE), and Mean Squared Error (MSE)



Hyperparameter Tuning

This section focuses on the importance of hyperparameters in determining how well the model can perform after tuning it to match the criteria. This underscores the importance of model optimization and fine-tuning in enhancing predictive accuracy.



Limitations and Further Research

This section discusses what were the limitations in this project and how further research into this project in the future can lead to further advancement in aerospace.

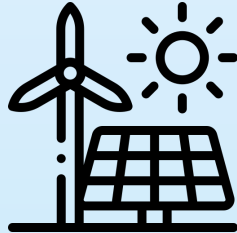
06

Future Applications →

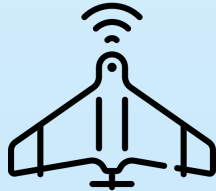




Applications



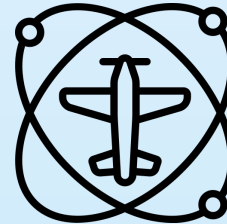
Renewable Industry



Unmanned Aerial Vehicles



Environment



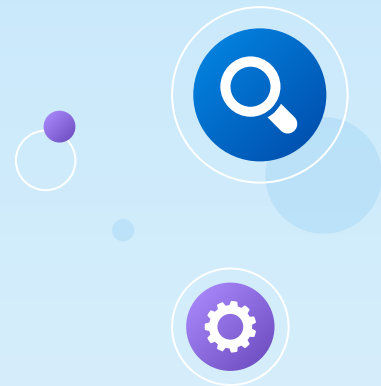
Aerospace Industry



Reliability

07

Conclusion →





Overview

This project utilized various ML models including:

- **Support Vector Regression**
- **Linear Regression**
- **Lasso Regression**
- **Ridge Regression**
- **Elastic-Net Regression**
- **Bayesian Ridge**
- **Random Forest Regression**
- **MLP**

to accurately classify the **airfoil** with the **most optimal lift-to-drag ratio** by predicting the most optimal lift-to-drag ratio based on **aerodynamic characteristics**.



Models

After initial models yielded an R^2 score below target, fine-tuning the model significantly improved its predictions. The final R^2 scores of **SVR** and **MLP** achieved was **0.86, 0.87**, respectively. Surpassing the project's target score of **0.8**, indicating accurate predictions.



Future

These results offer valuable insights into the design and efficiency of airfoils. With more accurate predictions of the lift-to-drag ratio, there is potential for advancements in areas such as aviation and wind energy.



Thank you

Q & A